

YEAR 12 ATAR PHYSICS
Units 3 and 4
ASSESSMENT OUTLINE 2020

ASSESSMENT TYPE	APPROXIMATE TIMING	ASSESSMENT WEIGHTING (%)	SCSA WEIGHTING (%)
Tests			
Task 2: Test on projectile and circular motion	Wk7_T4 2019	6	30
Task 3: Test on gravitational fields, satellites and moments	T1_Wk6	4	
Task 4: Test on electric fields, magnetic fields including the associated forces and energy	T1_Wk10	4	
Task 7: Test on electromagnetic induction	T2_Wk8	4	
Task 8: Test on wave particle duality and quantum theory	T3_Wk3	6	
Task 11: Test on special relativity and the standard model	T3_Wk9	6	
Examinations			
Task 6: Semester one exam	T2, Wk3/4	22	50
Task 12: Semester two exam	T3/4, (Oct holidays)	28	

Experiment Task 1: Circular motion experiment and validation Students will conduct an experiment where an object experiences a net force of constant magnitude, perpendicular to its velocity, resulting in the object undergoing uniform circular motion. A validation test will be completed at the end of this task. Task 5: Electromagnetism experiment and validation Students will conduct an experiment relating to forces experienced by charged objects in electric or magnetic fields. A validation test will be completed at the end of this task.	Wk7_T4 2019 T2_Wk1	2 6	20
Investigation Task 10: Standard Model Investigation Students will use information from various information sources to research and explain how the Standard Model can be used to explain the origins of the universe. A validation exercise will be completed at the end of task.	T3_Wk7	6	
Evaluation and analysis Task 9: Wave Particle Duality / Quantum Theory / Special Relativity Evaluation Students will research and evaluate evidence relating to Wave Particle Duality / Quantum Theory / Special Relativity. Students will construct scientific arguments relating to this evidence. A validation exercise will be completed at the end of the task.	T3_Wk4	6	

School-based assessment

The Western Australian Certificate of Education (WACE) Manual contains essential information on principles, policies and procedures for school-based assessment that needs to be read in conjunction with this syllabus.

Teachers design school-based assessment tasks to meet the needs of students. The table below provides details of the assessment types for the Physics ATAR Year 12 syllabus and the weighting for each assessment type.

Assessment table – Year 12

Type of assessment	Weighting
Science Inquiry There must be at least one experiment, one investigation and one evaluation and analysis completed in this pair of units. Appropriate strategies should be used to authenticate student achievement on an out-of-class assessment task. Experiment Practical tasks designed to develop or assess a range of laboratory related skills and conceptual understanding of physics principles, and skills associated with representing data; organising and analysing data to identify trends and relationships; recognising error, uncertainty and limitations in data; and selecting, synthesising and using evidence to construct and justify conclusions. Tasks can take the form of practical skills tasks, laboratory reports and short in-class tests to validate the knowledge gained. Investigation Activities in which ideas, predictions or hypotheses are tested and conclusions are drawn in response to a question or problem. Investigations can involve experimental testing, field work, locating and using information sources, conducting surveys, and using modelling and simulations. Assessment tasks can take the form of an experimental design brief, a formal investigation report requiring qualitative and/or quantitative analysis of the data and evaluation of physical information, or exercises requiring qualitative and/or quantitative analysis of second-hand data. Evaluation and analysis Involves interpreting a range of scientific and media texts; evaluating processes, claims and conclusions by considering the accuracy and precision of available evidence; and using reasoning to construct scientific arguments. Assessment tasks can take the form of answers to specific questions based on individual research; exercises requiring analysis; and interpretation and evaluation of physics information in scientific and media texts.	20%
Test Tests typically consist of questions requiring short answers, extended answers and problem solving. This assessment type is conducted in supervised classroom settings.	30%
Examination Examinations require students to demonstrate use of terminology, understanding and application of concepts and knowledge of factual information. It is expected that questions would allow students to respond at their highest level of understanding. Typically conducted at the end of each semester and/or unit and reflecting the examination design brief for this syllabus. This assessment type is conducted in supervised classroom settings.	50%